IS AI GOVERNABLE? INDUSTRY PERSPECTIVES ON THE ADOPTION, EFFECTIVENESS AND ACCOUNTABILITY OF FRONTIER AI

James Short
San Diego Supercomputer Computer, UC San Diego

June 2, 2025

This report is based on executive interviews and case analyses of 24 organizations, secondary analysis of company disclosures, investor and analyst briefings, research publications, industry white papers, case studies, and commentaries by technical, regulatory and policy experts. This manuscript is in the final stages of completion. It is being circulated to a selected panel of industry experts, researchers and academicians for review and comment. This version will undergo one further revision, where Acknowledgements, References, and the Appendix will be added, and the manuscript will then be formatted and issued as a SPARK AI Working Paper in June 2025.

A second version of this manuscript will be written for submission to a management journal over the summer. It will be revised and edited for length appropriately for journal submission, and incorporate selected quotes from the review panel.

PLEASE DIRECT ALL COMMENTS TO JAMES SHORT @ JSHORT@UCSD.EDU

CONFIDENTIAL DRAFT - PLEASE DO NOT COPY, DISTRIBUTE OR REFERENCE PUBLICLY

IS AI GOVERNABLE? INDUSTRY PERSPECTIVES ON THE ADOPTION, EFFECTIVENESS AND ACCOUNTABILITY OF FRONTIER AI

ABSTRACT

As artificial intelligence systems grow in scale, complexity, and capability, the question of how these systems can be effectively governed is increasingly urgent. This report presents a cross-sector analysis of how industry and public sector leaders are confronting the challenges of developing effective governance policies and ensuring accountability in frontier AI development. Drawing on insights from technical experts, policymakers and executives, it explores the emerging infrastructure, standards, and mechanisms shaping the governance landscape.

INTRODUCTION

Artificial Intelligence (AI) can no longer be considered an emerging technology — it is a defining force in contemporary business, government and global IT adoption and governance. As its diffusion and application accelerate across sectors, so too does the imperative to understand the feasibility, mechanisms, and implications of governing it.

This report examines four critical decision points that leaders must navigate in defining effective AI governance. They are based on executive interviews and case studies in over two dozen organizations, and secondary analysis of hundreds of announcements and disclosures made by producers, consumers, analysts and commentators on the evolving AI governance ecosystem. The study methodology is outlined in the Appendix.

Is AI Governable?

First, we ask a foundational question: Is AI governable? How can we measure the effectiveness of governance policies? Much of today's AI — particularly machine learning and generative models — behave less like a static tool and more like a probabilistic system that evolves over time and adapts to changing inputs. This raises novel challenges for accountability, auditability, control and effectiveness. Unlike traditional software, AI outputs may not be fully predictable, explainable, or traceable, even to their developers. AI systems often integrate across global supply chains and data ecosystems, making jurisdictional oversight complex. Moreover, different organizations will have different assumptions of what governance is, and what it should be. What does governance mean to you? Are current technical governance mechanisms and maturity models adequate to the task? Finally, are these truly new problems, or familiar ones intensified by AI's speed, scale and scope? Before overhauling governance models entirely, we must first assess how current legal, institutional, and technical tools fall short — and where they may still be fit for purpose.

Who Governs AI?

Governance begins with the question of who governs AI? Which depends on the company and individual's frame of reference. Respondents in our study identified over 20 different stakeholder roles across technology, management, risk assessment, compliance, and governance functions—ranging from CIOs and CTOs to corporate boards and advisory councils. Many organizations are experimenting with

intermediating structures such as AI centers of excellence or governance councils that include cross-functional stakeholders. However, the effectiveness and legitimacy of these mechanisms are not guaranteed. In practice, corporate actors often wield disproportionate influence over governance design and decision-making, raising concerns about bias, effectiveness and public accountability. This section explores both procedural questions—such as who sets the agenda in multi-stakeholder forums—and substantive concerns about how influence and authority are distributed, and balanced, in governing AI.

Does Strong Governance Inhibit Innovation?
Third, we examine whether governance and innovation are allies or adversaries. A central argument in this long-running debate suggests that meaningful oversight can only be introduced after an innovation has been widely adopted and market-tested — not before. The fear is that premature regulation risks locking in assumptions, limiting experimentation, or stifling investment in uncertain or emerging areas. However, history shows that this trade-off is not always inevitable. Smart regulation — crafted in collaboration with innovators and calibrated to risk — can work together to build trust, reduce uncertainty, and in the best case, accelerate adoption. The challenge is distinguishing between governance that enforces necessary guardrails and governance that becomes a bottleneck. In the AI context, we ask: is the tension between innovation and regulation overstated? Or are we using it as a rationale to delay action that is already overdue?

What is the Economic Cost of Governance?
Fourth, we spotlight an often-overlooked issue: the economic cost of governance. While responsible and strategic imperatives for AI oversight are frequently emphasized, there is little concrete analysis of what governance actually entails in financial terms. Compliance costs can include everything from regulatory reporting, audit mechanisms, and model documentation to third-party assessments, secure data handling, and redress procedures — all of which impose direct and ongoing financial burdens on organizations. Opportunity costs are subtler but potentially more significant: the delay or abandonment of high-value deployments due to uncertain regulatory landscapes or preemptive self-censorship. And then there is institutional capacity — the often-underestimated need for talent, training, and organizational redesign required to effectively implement and monitor governance frameworks, both within firms and across regulatory bodies. Without a clearer understanding of these costs, we risk building governance structures that are either economically unsustainable or strategically misaligned.

IS AI GOVERNABLE?

The question of whether artificial intelligence (AI) is governable elicited both optimism and caution among our interview participants. Respondents emphasized that, because AI systems are designed and developed by humans, robust institutional, legal, and technical controls are necessary. Human involvement was seen not as a guarantee of reliability, but also a potential source of error—introducing unforeseen bias, employing flawed assumptions, or creating unintended consequences. Respondents highlighted the accelerating speed, scale, and complexity of AI development as key challenges to existing oversight mechanisms. Concerns were raised about whether governance tools and organizational structures are keeping pace with technological change. Internal misalignments between technical implementation and governance practices were noted as common and potentially consequential. Looking ahead, several participants pointed to the emergence of machine-generated AI systems—where AI designs other AI (e.g., via neural architecture search or AutoML)—as a development

that introduces new policy dilemmas. These systems may reduce transparency into how models are created, complicating efforts to trace decisions or ensure accountability.

Governance Models and Frameworks
Companies in our sample referenced emerging AI governance maturity models—such as the EU Artificial Intelligence Act and the OECD AI Principles—as sources of high-level guidance. Technical frameworks such as the NIST AI Risk Management Framework (RMF) were also seen as providing structured, procedural tools for identifying, assessing, and mitigating risks. Still, respondents noted that such frameworks had to evolve dynamically with the technology to remain effective.

Critical Policy Drivers
Four key policy drivers emerged as central to shaping AI governance practices:

Risk Tolerance
Risk tolerance was defined as the acceptable level of deviation from performance goals that an organization is willing to accept in pursuit of its strategic objectives. Respondents emphasized the role of corporate risk culture in shaping how senior management evaluates and manages the uncertainties of AI adoption, including potential security threats, system failures, regulatory ambiguity, and reputational harm.

Regulatory Foresight
Proactive anticipation of future regulatory developments—especially in heavily regulated sectors—was viewed as essential. Many companies had developed or were developing formal mechanisms to model and test potential regulatory scenarios through tools like policy sandboxes and pilot programs. Opinions diverged on the sufficiency of self-regulation, with over half of the participants arguing that industry-led frameworks, in the absence of legal enforcement, often fall short due to regulatory gaps, weak accountability, limited enforcement power and legal risk.

Innovation Incentives
A company's innovation culture and the incentives provided to line managers were seen as critical to driving AI adoption. External motivators included competitive pressure, reputational gains, and potential regulatory relief. Internally, respondents emphasized the importance of change management, particularly in addressing workforce concerns that AI may replace rather than augment their roles. Successful governance was seen to depend, in part, on how effectively senior leadership engaged and upskilled employees in adapting to new technologies.

Cost Tolerance
Most organizations lacked a clear approach to assessing or managing the financial and operational costs of AI governance. The concept of cost tolerance—defined as an organization's capacity and willingness to invest in responsible AI practices—was often underdeveloped. Few companies had formal budgeting processes, resource allocation plans, or cost-benefit frameworks dedicated to governance. This gap presents a significant opportunity: by establishing clearer thresholds and investment strategies, organizations can make more strategic decisions about when, where, and how to embed responsible AI practices at scale.

The following exhibit summarizes key differences in policy responsibility, authority and accountability among governance, compliance and control mechanisms.

Exhibit 1

GOVERNANCE	COMPLIANCE	CONTROL
Strategic	Legal/Regulatory Adherence	Operational Mechanisms
Proactive Oversight	Reactive Enforcement	Tactical Implementation
Responsible, Accountable Direction	Mandatory Obligations	Preventative Safeguards, Guardrails
Example: Board, Corporate Governance Policies	Examples: GDPR, HIPAA, SOX Compliance	Example: Segregation of Functions, Access Controls

WHO GOVERNS AI?

AI governance begins with a fundamental - often contested - question: who governs AI? The answer varies widely across industries, organizations, and management roles. Our interviews reported that AI governance is not the responsibility of a single organizational function, group or leader ("czar"), but is unfolding within a complex, frequently fragmented, landscape of stakeholders with differing responsibilities, interests, and influence.

Role Fragmentation

Survey respondents identified more than 20 distinct roles involved in AI governance, spanning technical, operational, legal and strategic responsibilities. These included traditional leadership roles such as Chief Information Officers (CIOs), Chief Technology Officers (CTOs), Chief Risk Officers, and Chief Compliance Officers, but also extended to legal teams, cybersecurity, data science, HR, line management, and external advisory boards. Several companies had established cross-functional councils or centers of excellence, under different names and with varied scopes. However, their primary functions were similar—providing cross-functional oversight, reviewing high-risk use cases, and establishing internal policies to align AI deployment with risk assessment, security, compliance and regulatory standards.

Intermediating Governance Structures

To manage this complexity, most of the organizations we interviewed were establishing intermediating structures aimed at improving coordination and shared oversight. Structures included AI Centers of Excellence, Ethics Review Boards, and/or AI Governance Councils charged with convening cross-functional teams to deliberate the development of key governance policies. These bodies typically operated using collaborative models—ranging from consultative to participatory to co-creation-based approaches.

When well-resourced and supported, these structures can play important roles in balancing competing interests and priorities across business units. However, survey participants stated that their effectiveness was far from guaranteed. Many of these bodies are purely advisory, lacking formal authority, decision rights, or operational influence. Without adequate resources, clear mandates, and executive sponsorship, these mechanisms risk becoming more symbolic than actionable.

Information and Influence Asymmetries
As AI systems increasingly underpin public-facing applications—such as hiring, lending, healthcare, or law enforcement—internally focused governance policies may prove insufficient. Without meaningful public accountability mechanisms, there is the risk that governance remains narrowly tailored to company interests, failing to address broader civic or societal concerns such as transparency, accountability and public trust.

The importance of external oversight to address these asymmetries surfaced in several interviews. Oversight mechanisms cited included formal regulation, civic engagement, ombudsman functions, or independent third-party audits.

Procedural Governance Challenges
AI governance is shaped also by how decisions are made. Respondents noted the procedural importance of agenda-setting influence within multi-stakeholder forums—specifically, who defines the problem, who frames the risks, and who decides what constitutes success or failure. Substantively, these challenges are not unique to AI, they are present in varying degrees in all disruptive technologies. However, a common response in our sample was the scale, speed and scope of AI heightened their urgency.

DOES STRONG GOVERNANCE INHIBIT INNOVATION?

A key decision in defining corporate AI governance policy involves how senior management balances strategic objectives, competitive positioning, risk tolerance, and financial structuring. Regulatory expectations help define the boundary conditions for where oversight mechanisms—such as internal guardrails—can be implemented, monitored, and enforced. These conditions influence where and how regulators expect firms to ensure compliance, and where, in the absence of such oversight, companies have greater discretion to innovate—whether in strategic direction, product and service development, operational design, or financial modeling.

In principle, innovation, regulation, and compliance can be balanced to create fit-for-purpose AI governance policies. However, the majority of survey participants indicated that this balance has been disrupted by rapid technical advancement in AI, regulatory uncertainty, lack of operational experience with production-scale systems, limited cost visibility, and the absence of stakeholder consensus. As one respondent noted, *"we are still very early in this process."* These tensions are intensified in fast-moving technological domains like AI, where conflicting policy drivers and trade-offs are more acute. As investors, developers, and vendors may view early-stage governance efforts as overly restrictive—particularly when these efforts introduce compliance burdens in the absence of clear legal mandates or when they see standards evolving more slowly than the underlying technology. In such cases, regulation can function more as a brake than a guide.

Mixed Evidence
The historical record is decidedly mixed as to whether regulation accelerates or hinders innovation. In some sectors—such as aviation, pharmaceuticals, and financial services—robust governance frameworks were introduced early or alongside key innovations. And in these cases, regulation helped establish standardized protocols, reduced systemic risks, and = perhaps most critically – created policy

incentives and controls advancing interoperability across markets. In broader economic terms, these conditions are frequently cited as drivers of market expansion, economic growth and innovation. Conversely, sectors such as healthcare, energy, small business, and the entertainment industry have long complained that overly burdensome, inflexible, or outdated regulation have constrained innovation and suppressed market potential.

In the context of AI, several key industry players have advocated that companies—starting with their own—should develop and implement internal AI self-governance frameworks. In the absence of comprehensive AI-specific legislation in the United States, many argue that firms should neither proceed without controls nor wait passively for regulatory clarity. Over half of respondents in our survey agreed on the importance of defining self-governance structures as a core component of the broader AI governance process.

Self-Governance

Self-governance refers to the voluntary development and enforcement of rules, standards, and accountability mechanisms by the companies themselves. In practice, self-governance responsibilities are usually distributed among legal (CLO), risk (CRO), and compliance functions. Respondents, as well as supporting literature, suggest that effective self-governance frameworks should be grounded in four principles:

1. *Clear delineation of roles and accountability across the organization—including boards, senior business leaders, and technical executives.*
2. *Explicit standards and principles for evaluating, monitoring, and assessing AI system behavior and performance.*
3. *Data privacy impact assessments for all data collected by or shared with AI systems, to ensure compliance with applicable privacy and security laws.*
4. *Assignment of contractual responsibilities and liabilities between developers, vendors, and acquirers of AI systems through legally binding agreements.*

However, a number of obstacles hinder the development, effectiveness and accountability of self-governance mechanisms, and externally, public acceptance and trust in self-governance policies. Obstacles include misaligned incentives, lack of internal expertise, absence of control and enforcement mechanisms, and unclear ownership and of self-governance functions. This interview question prompted a variety of perspectives on the responsibility and effectiveness of self-governance policies, highlighting the distinction between compliance practices that were long established, and governance practices that were seen as early in maturity. Exhibit 2 presents a summary of the perspectives we heard. With the caveat that the table represents an aggregate of perspectives, some based on direct experience with new governance functions that were in place, and some viewed as design considerations in defining new governance policies in deliberation.

Exhibit 2. Effective vs. Ineffective Practices in AI Self-Governance

Policy Dimension	Seen As Effective Practice	Seen As Ineffective Practice
Transparency	The commitment to publicly disclose AI policies, model capabilities, limitations, known and potential risks.	Selective information made available on model design, usage and risks.
Accountability Mechanisms	Uses internal audits, third-party reviews, defines and enforces consequences for violations.	Lacks clear ownership and accountability for risk assessment, and consequences for harmful outcomes.
Risk Assessment	Implements structured, repeatable risk evaluations, including pre-deployment reviews to help define risk potential across the AI development lifecycle.	Risk management is ad hoc or implicitly reactive; process lacks formal assessment procedures and oversight.
Stakeholder Engagement	Commitment to broad stakeholder engagement, Involving domain experts, business stakeholders, and iwhere civic society may be affected, participation in the governance process	Organized as an internal governance function with limited input from external or potentially impacted stakeholders.
Standards Alignment	Commitment to align, with oversight, to civic, industry and public norms. For example, NIST, ISO, OECD AI principles.	No clear commitment to alignment with civic, industry and public governance norms.
Adaptability	Views governance policy as evolving, needing continual assessment. Engages stakeholders for periodic review, especially as it concerns incident reporting and response.	Maintains static frameworks despite evolving risks or market expectations.
Motivations	Drivers include long-term strategy, adaptability of potential risk categories driven by AI technical progress and adoption by business partners, customers and competitors, variables important in ecosystem and market stability.	Motivated primarily by narrow risk assessment and reputational management, lobbying, or optics.

CONFIDENTIAL DRAFT - PLEASE DO NOT COPY, DISTRIBUTE OR REFERENCE PUBLICLY

THE ECONOMICS OF GOVERANCE:
NAVIGATING FROM COST CENTER TO STRATEGIC INVESTMENT

As AI capabilities and risk assessment methodologies become more deeply embedded across the enterprise, governance will evolve from compliance and risk assessment to an essential enabler of both strategic execution and risk management. For top management and corporate boards, the core question is not *whether* to invest in AI governance, but how to calibrate that investment to match the company's strategic ambitions, risk tolerance and resource allocation. The governance function will need adequate resources to ensure compliance and risk, but also to ensure the broader strategic goals of AI deployment velocity, business value realization, stakeholder trust, and the roadmap for scaling institutional readiness for future AI.

While the great majority of our sample companies were committed to, and actively pursuing lateral coordination functions to ensure compliance and risk assessment, a few of the companies and participants were seeing the need for the next step - "readiness and preparation for next generation AI governance" - as one interview participant stated it. Such a shift would involve incorporating a more strategic, fiduciary-aligned vision addressing the company's top management and board mandate: balancing value creation, risk oversight, and resource allocation:

- Preserving value creation by ensuring governance does not become a bottleneck to high-potential AI deployments.
- Containing downside risk by meeting rising regulatory obligations and preventing reputational or compliance failures.
- Scaling internal readiness by building institutional capacity — talent, tooling, and decision making roles — aligned with AI assessment, approval, deployment, and monitoring

With the above as context, a deliberate, calibrated investment approach would focus spending in three interconnected cost domains: compliance infrastructure, opportunity cost mitigation, and institutional capability building. Without commitments in each, governance frameworks may risk becoming either economically unsustainable or strategically misaligned.

Compliance Infrastructure: Meeting a Moving Regulatory Target

Governance investments must first ensure the company's ability to meet baseline regulatory and accountability requirements. This includes model documentation, audit trails, third-party assessments, and secure data handling protocols — all of which have direct cost implications in both technology and staffing. As AI regulation becomes more formalized (e.g., EU AI Act, U.S. sectoral rulemaking), compliance will increasingly require dedicated cross-functional teams and budgeted reporting capabilities. For CFOs, this domain may be viewed as a form of regulatory insurance — the cost of which rises with model complexity and geographic exposure.

Opportunity Cost Mitigation

Second, insufficient governance can result in avoidable delays, abandoned initiatives, or excessive internal risk aversion. When deployment timelines slip due to unclear guardrails or unresolved accountability, costs can include foregone revenue, lost market timing, and loss of competitiveness. The

goals of strategic investment in AI governance is to reduce these costs by providing clear decision roles, pre-approved deployment protocols, and where appropriate, fast-track review mechanisms for lower-risk use cases. In this case, the CFO lens is ROI-protective: a modest upfront investment can preserve the economics of larger, more strategic digital efforts.

Institutional Capability Building: Making Governance Sustainable

Third, AI governance must be implemented through capable staff, repeatable processes, and scalable processes. These require investments in AI fluency for relevant teams (legal, audit, risk), role clarity across functions, and the development and/or procurement of software to support model inventorying, version control, and risk scoring. These functions are often underestimated line items in early governance programs, yet are critical for long-term resilience. For top management and the board, this is where governance evolves from a compliance function into a strategic control system. One that grows with the company's AI ambitions and footprint.

Sustainable AI Governance

Viewed in isolation, AI governance can appear as overhead. But when mapped against the risks and costs involved in AI project failure, regulatory sanction, or strategic inaction, governance can be viewed financially as a protective investment — a way to navigate the upside and downside of AI. Boards and CFOs increasingly will expect governance functions to come with defined financial profiles, tracked against performance metrics such as deployment velocity, audit readiness, and internal alignment. This approach reframes governance as a capital allocation decision: boards and CFOs are not simply funding oversight, but underwriting the conditions for sustainable innovation. From this perspective, governance merits financial planning and performance monitoring alongside other critical enablers of AI transformation.

FUTURE RESEARCH: VALIDATING THE KNOWN, INVESTIGATING THE UNKNOWN

Any discussion of AI governance is inherently a forward-looking research topic. This brief has identified four decision scenarios identified in governance surveys and secondary research. If we rerun our survey in six months, it is likely that while our decision scenarios will still be recognizable, but our answers and insights may be different, perhaps predictably so, or perhaps not. The pace of AI development may be moving faster than the pace of our understanding of AI. Laws and policy invariably lag technological advancement. AI is starting to reshape labor markets, company competitiveness, and more broadly in education, art, and civic communications. Many institutions and people are not prepared for this transition.

On the other hand, AI is already improving healthcare diagnostics, scientific research, knowledge and information accessibility, and more. Delaying progress could postpone solutions to urgent problems. And in many cases, real-world deployment is necessary to uncover problems that lead to better thinking, tools, norms, and policies.

<working on the last paragraph, topics and issues for future research>

PENDING SECTIONS TO COMPLETE
WILL FINALIZE IN PARALLEL TO DISTRIBUTING THIS VERSION FOR REVIEW AND COMMENT

ACKNOWLEDGEMENTS (IN MANY CASES REQUIRES PERMISSION TO NAME)

REFERENCES (LIMITED TO ONE PAGE)

APPENDIX (ONE PAGE ON STUDY OVERVIEW AND METHODS)

<END OF MANUSCRIPT>